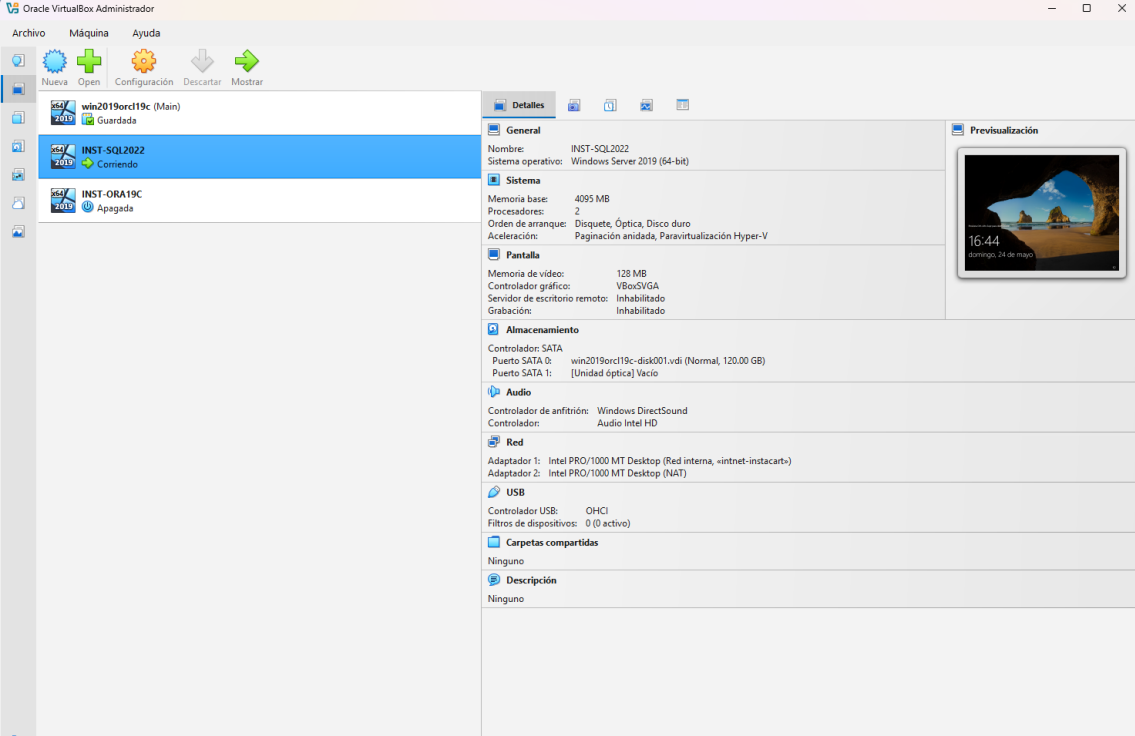
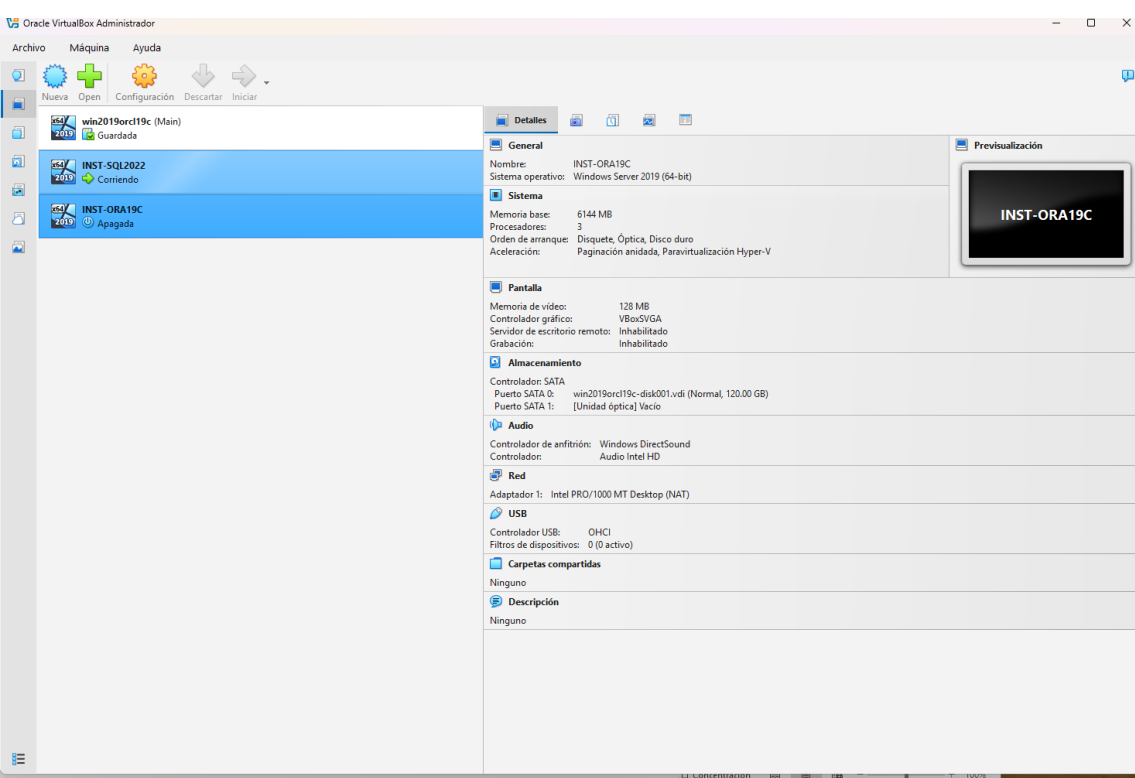


Proceso de virtualización:

Maquina virtual den Oracle Virtual Box INST-SQL:

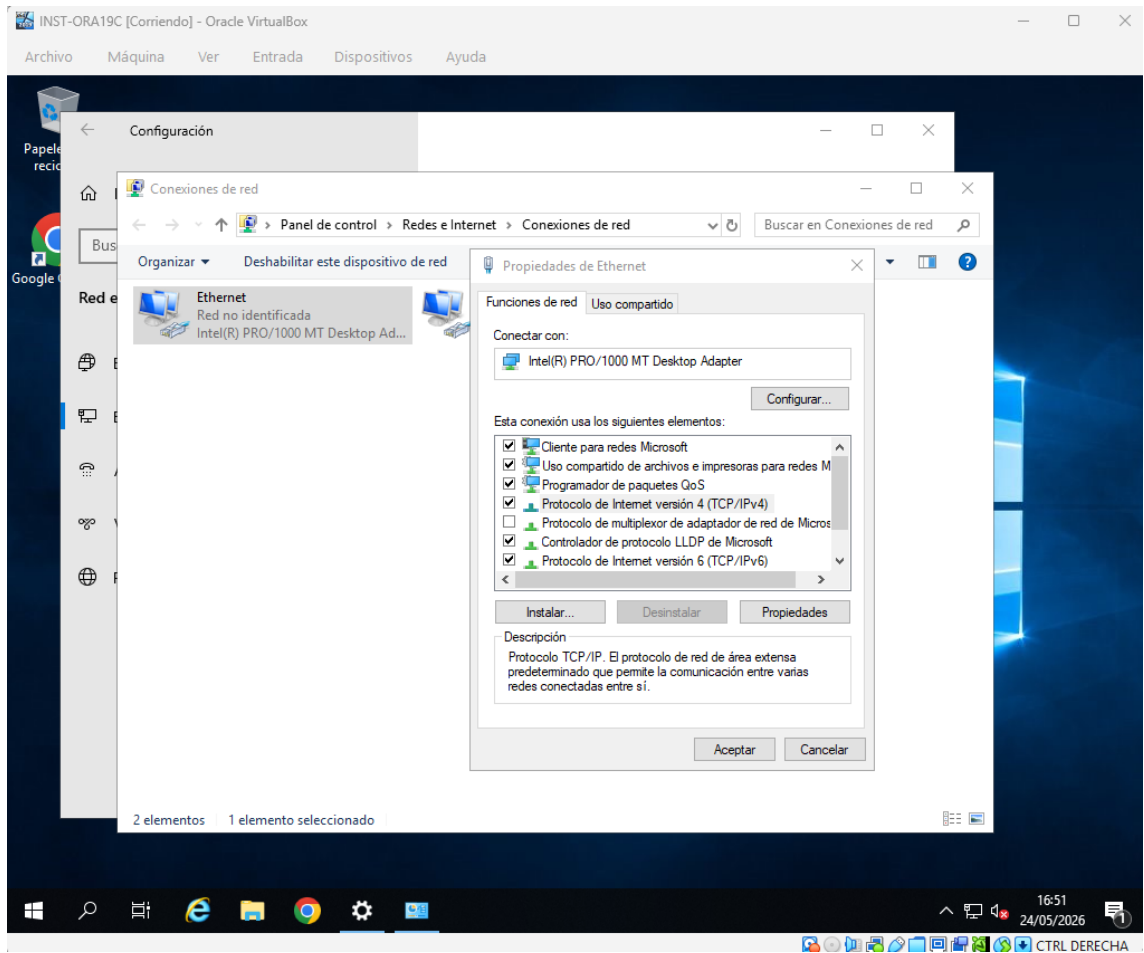


Maquina virtual den Oracle Virtual Box INST-ORA19:

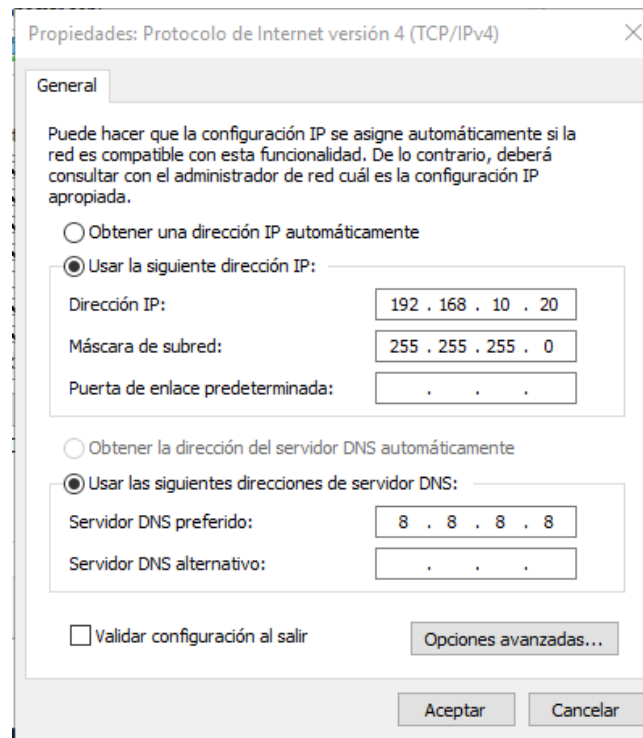


Configuracion de IPS para Red Interna:

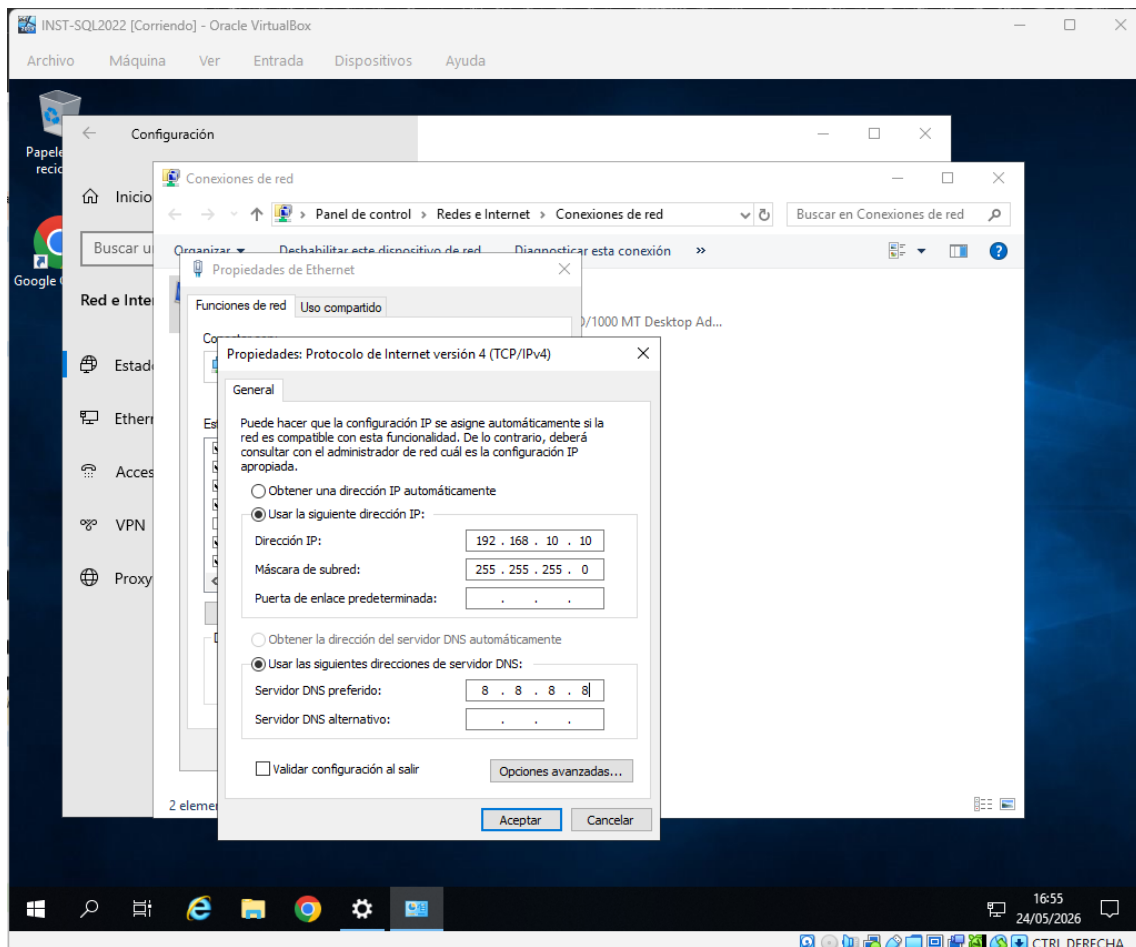
INST-ORA:



IPS:

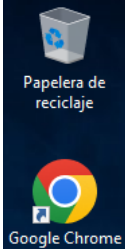


INST-SQL:



PRUEBAS DE CONEXIÓN DE IPS EN RED INTERNA DE LAS VM

ORA:



Administrador: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

PS C:\Users\Administrador>
PS C:\Users\Administrador> netsh advfirewall set allprofiles state on
Aceptar
PS C:\Users\Administrador> ping 192.168.10.10

Haciendo ping a 192.168.10.10 con 32 bytes de datos:
Respuesta desde 192.168.10.10: bytes=32 tiempo=1ms TTL=128
Respuesta desde 192.168.10.10: bytes=32 tiempo<1m TTL=128
Respuesta desde 192.168.10.10: bytes=32 tiempo<1m TTL=128
Respuesta desde 192.168.10.10: bytes=32 tiempo<1m TTL=128

Estadísticas de ping para 192.168.10.10:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
              (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 0ms, Máximo = 1ms, Media = 0ms
PS C:\Users\Administrador> netsh advfirewall show allprofiles state

Configuración de Perfil de dominio:
-----
Estado                                ACTIVAR

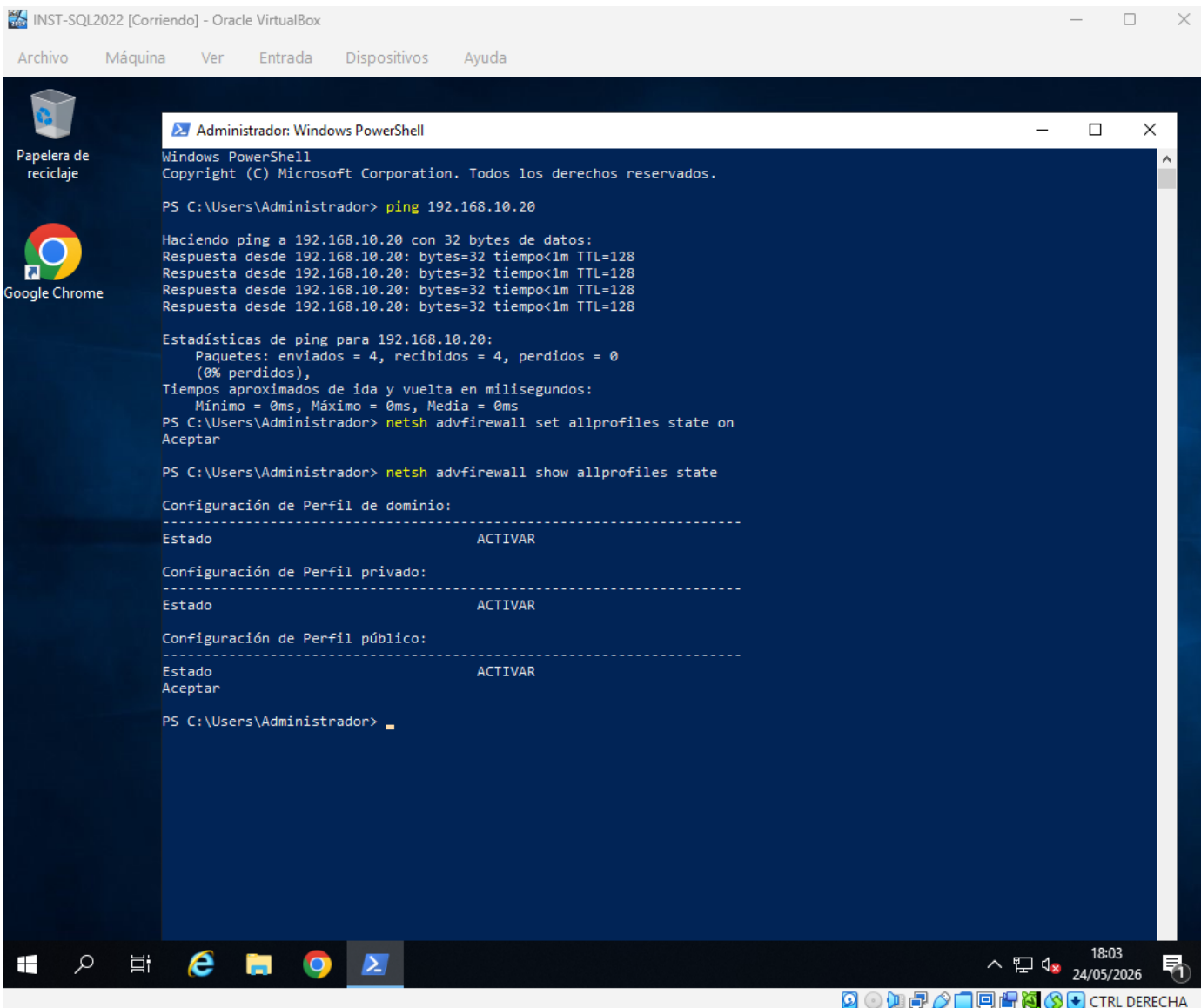
Configuración de Perfil privado:
-----
Estado                                ACTIVAR

Configuración de Perfil público:
-----
Estado                                ACTIVAR
Aceptar
```

```
PS C:\Users\Administrador>
```



SQL:



```
INST-SQL2022 [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

Papera de reciclaje
Google Chrome

Administrador: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

PS C:\Users\Administrador> ping 192.168.10.20

Haciendo ping a 192.168.10.20 con 32 bytes de datos:
Respuesta desde 192.168.10.20: bytes=32 tiempo<1m TTL=128
Respuesta desde 192.168.10.20: bytes=32 tiempo<1m TTL=128
Respuesta desde 192.168.10.20: bytes=32 tiempo<1m TTL=128
Respuesta desde 192.168.10.20: bytes=32 tiempo<1m TTL=128

Estadísticas de ping para 192.168.10.20:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
              (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 0ms, Máximo = 0ms, Media = 0ms
PS C:\Users\Administrador> netsh advfirewall set allprofiles state on
Aceptar

PS C:\Users\Administrador> netsh advfirewall show allprofiles state

Configuración de Perfil de dominio:
-----
Estado                                ACTIVAR

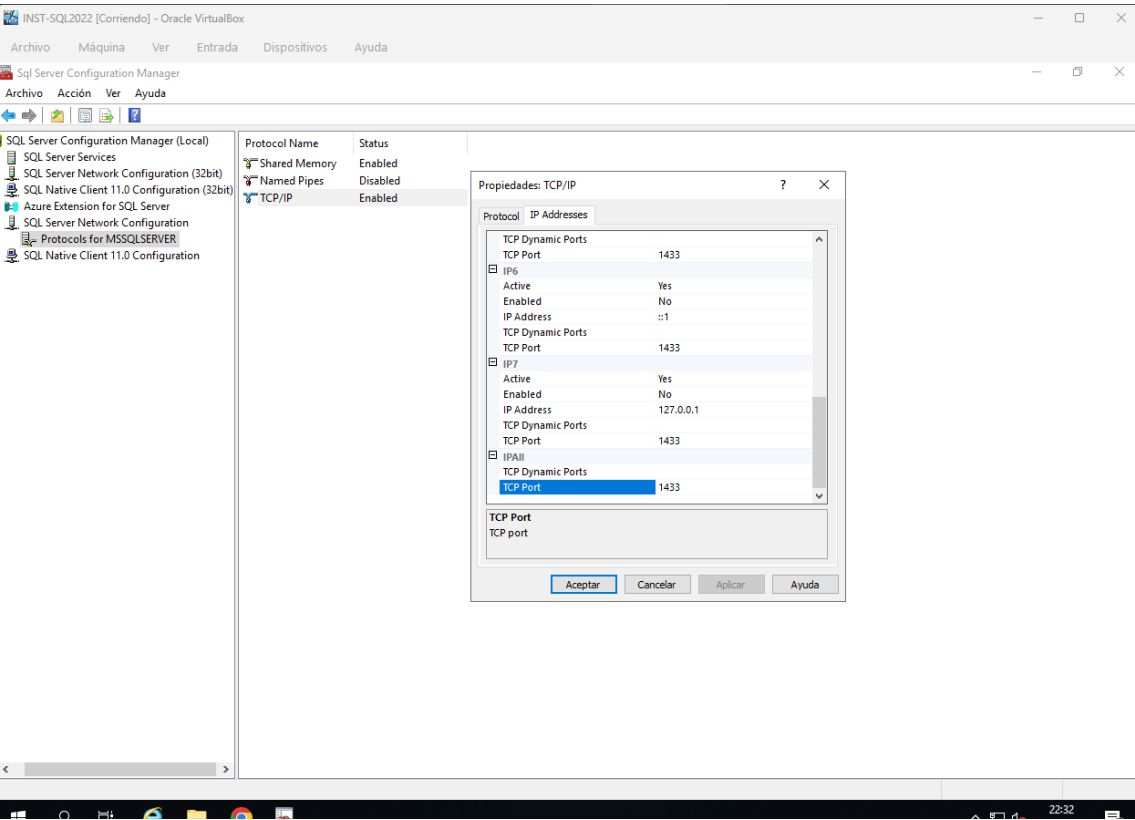
Configuración de Perfil privado:
-----
Estado                                ACTIVAR

Configuración de Perfil público:
-----
Estado                                ACTIVAR
Aceptar

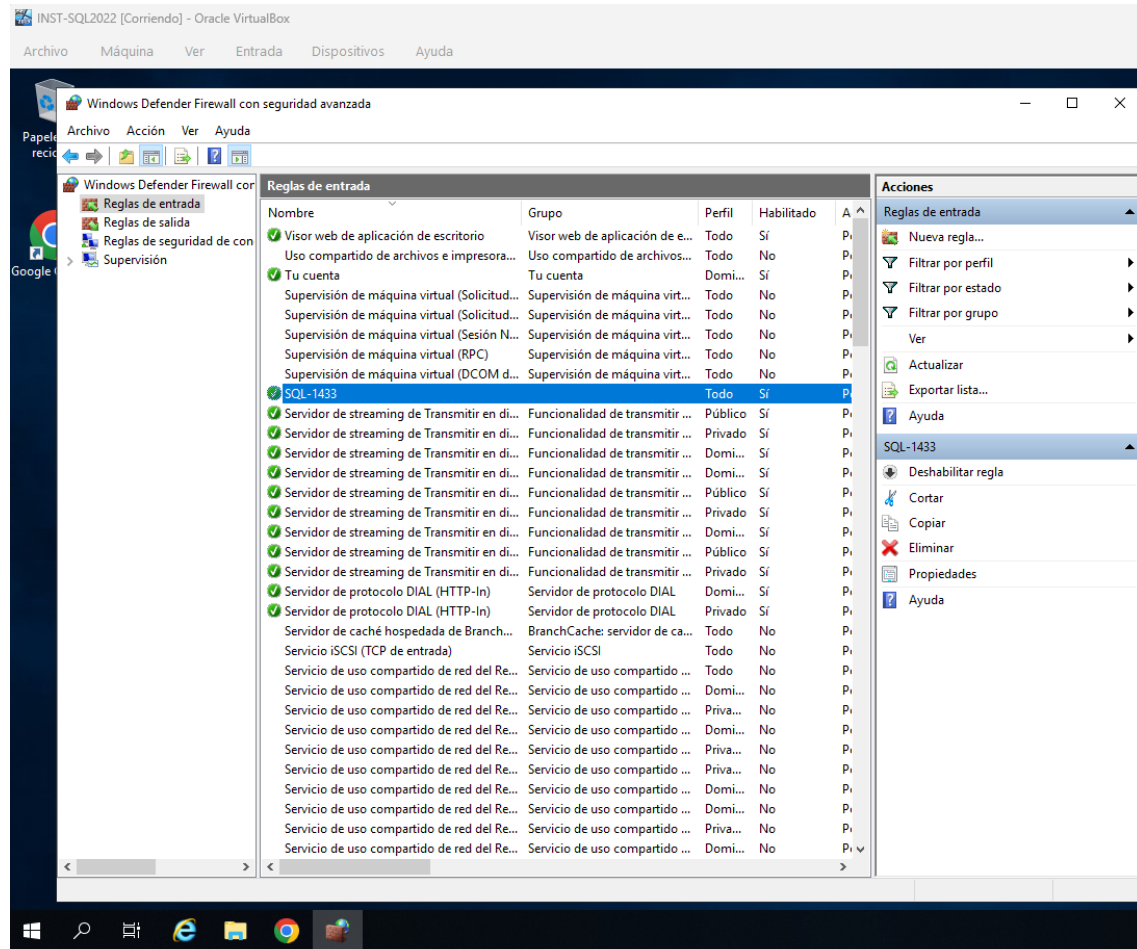
PS C:\Users\Administrador>
```

18:03
24/05/2026
CTRL DERECHA

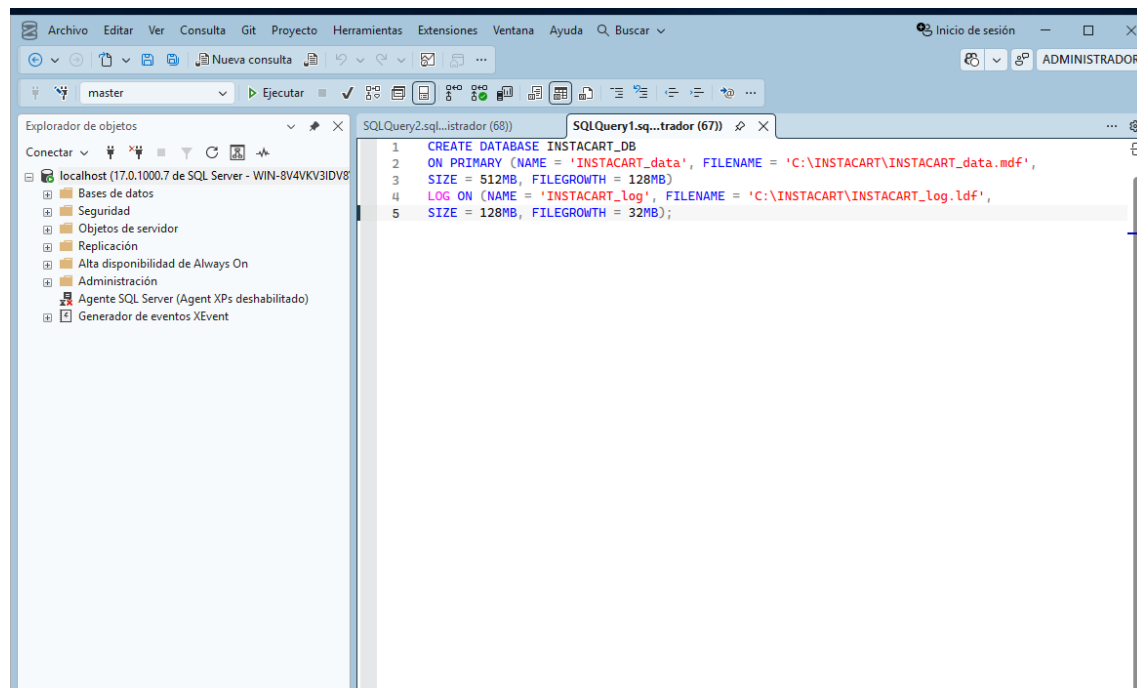
HABILITANDO TCP/IP

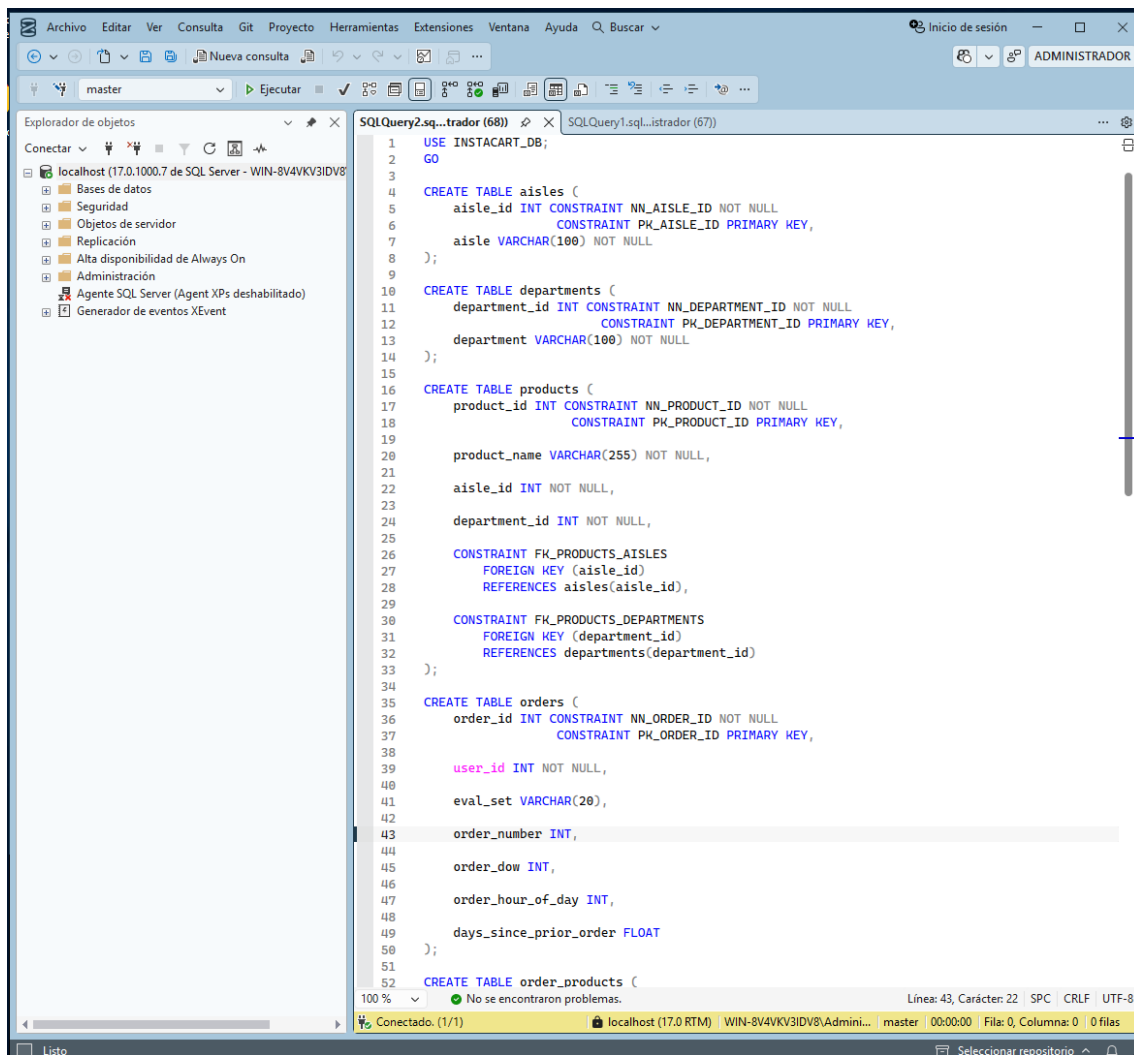


FIREWALL NUEVA REGLA:



CREAMOS LA DATABASE EN SQL CON SUS TABLAS PARA TRANSFERIR
LOS DATOS DE LA EMPRESA INSTACARD A NUESTRO SQL:





```
CREATE TABLE order_products (
    order_id INT NOT NULL,

    product_id INT NOT NULL,

    add_to_cart_order INT,

    reordered BIT,

    CONSTRAINT PK_ORDER_PRODUCTS
        PRIMARY KEY (order_id, product_id),

    CONSTRAINT FK_OP_ORDERS
        FOREIGN KEY (order_id)
        REFERENCES orders(order_id),

    CONSTRAINT FK_OP_PRODUCTS
        FOREIGN KEY (product_id)
        REFERENCES products(product_id)
);
```

BULKS:
Aisles

The screenshot displays two SQL queries in the SQL Server Enterprise Manager interface. The first query, in the 'SQLQuery1.sql...istrador (67)' tab, performs a bulk insert into the 'aisles' table and then counts the records. The second query, in the 'SQLQuery2.sql...istrador (68)' tab, performs a bulk insert into the 'departments' table, counts the records, and then performs another bulk insert into the 'products' table, which also includes a count. The results pane at the bottom shows a single row for the 'products' table with a count of 49688.

```
SQLQuery1.sql...istrador (67)
1  USE INSTACART_DB;
2  GO
3
4  BULK INSERT aisles
5  FROM 'C:\INSTACART\csv\aisles.csv'
6  WITH (
7      FIRSTROW = 2,
8      FIELDTERMINATOR = ',',
9      ROWTERMINATOR = '0x0a'
10 );
11
12 SELECT COUNT(*) AS aisles FROM aisles;
13
```

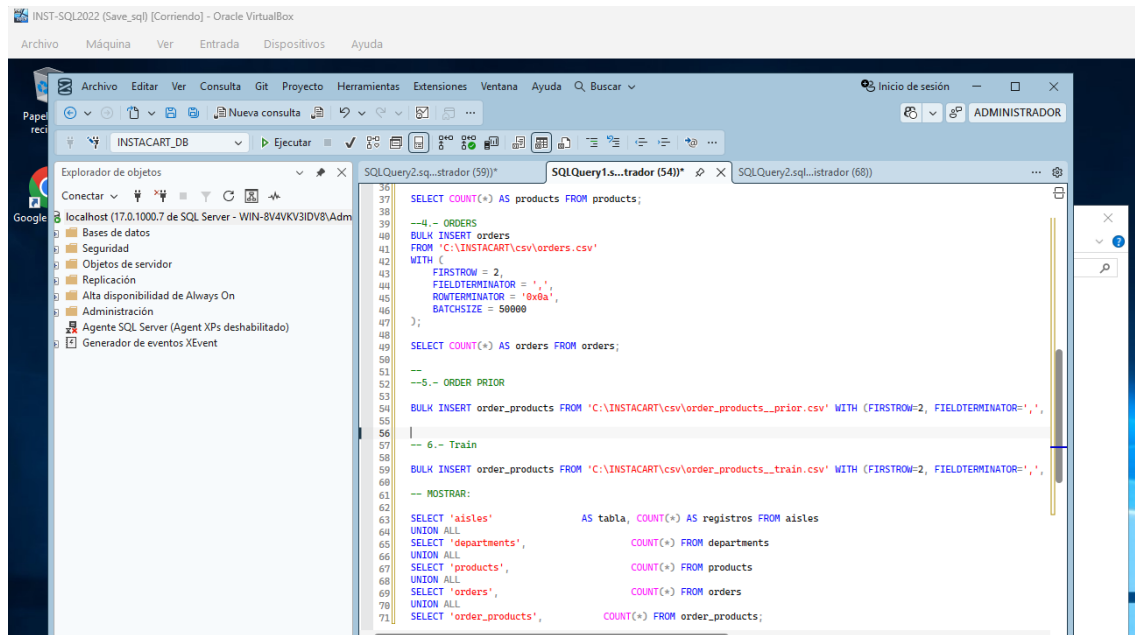
```
SQLQuery2.sql...istrador (68)
14 -- 2. departments
15 BULK INSERT departments
16 FROM 'C:\INSTACART\csv\departments.csv'
17 WITH (
18     FIRSTROW = 2,
19     FIELDTERMINATOR = ',',
20     ROWTERMINATOR = '0x0a'
21 );
22
23 SELECT COUNT(*) AS departments FROM departments;
24 GO
```

```
SQLQuery1.sql...istrador (67)
22
23 SELECT COUNT(*) AS departments FROM departments;
24 GO
25
26 --3.- Productos
27
28 BULK INSERT products
29 FROM 'C:\INSTACART\csv\products_clean.csv'
30 WITH (
31     FIRSTROW = 2,
32     FIELDTERMINATOR = ',',
33     ROWTERMINATOR = '0x0a',
34     MAXERRORS = 100
35 );
36
37 SELECT COUNT(*) AS products FROM products;
```

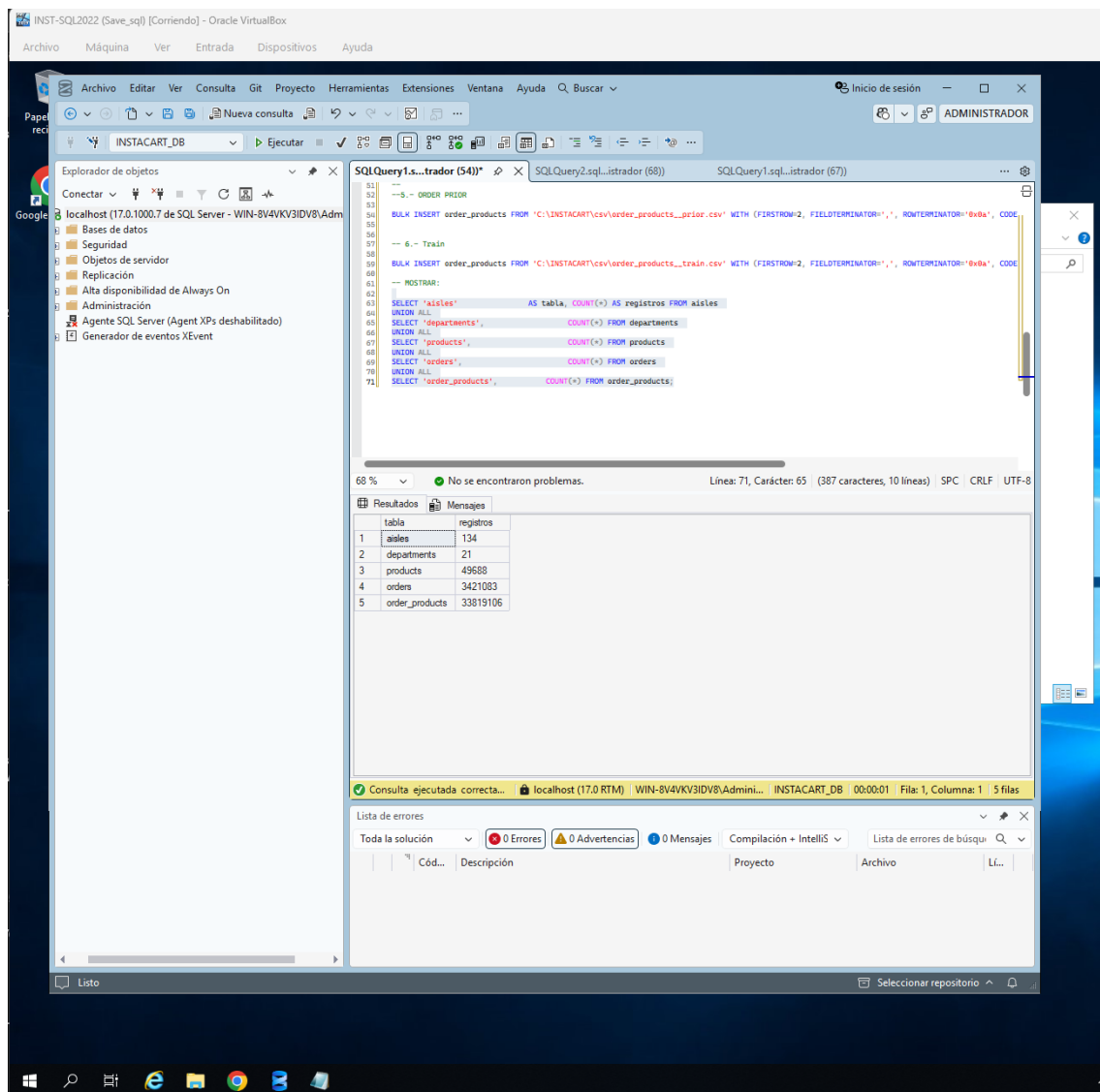
100 % No se encontraron problemas. Línea: 36, Car

Resultados Mensajes

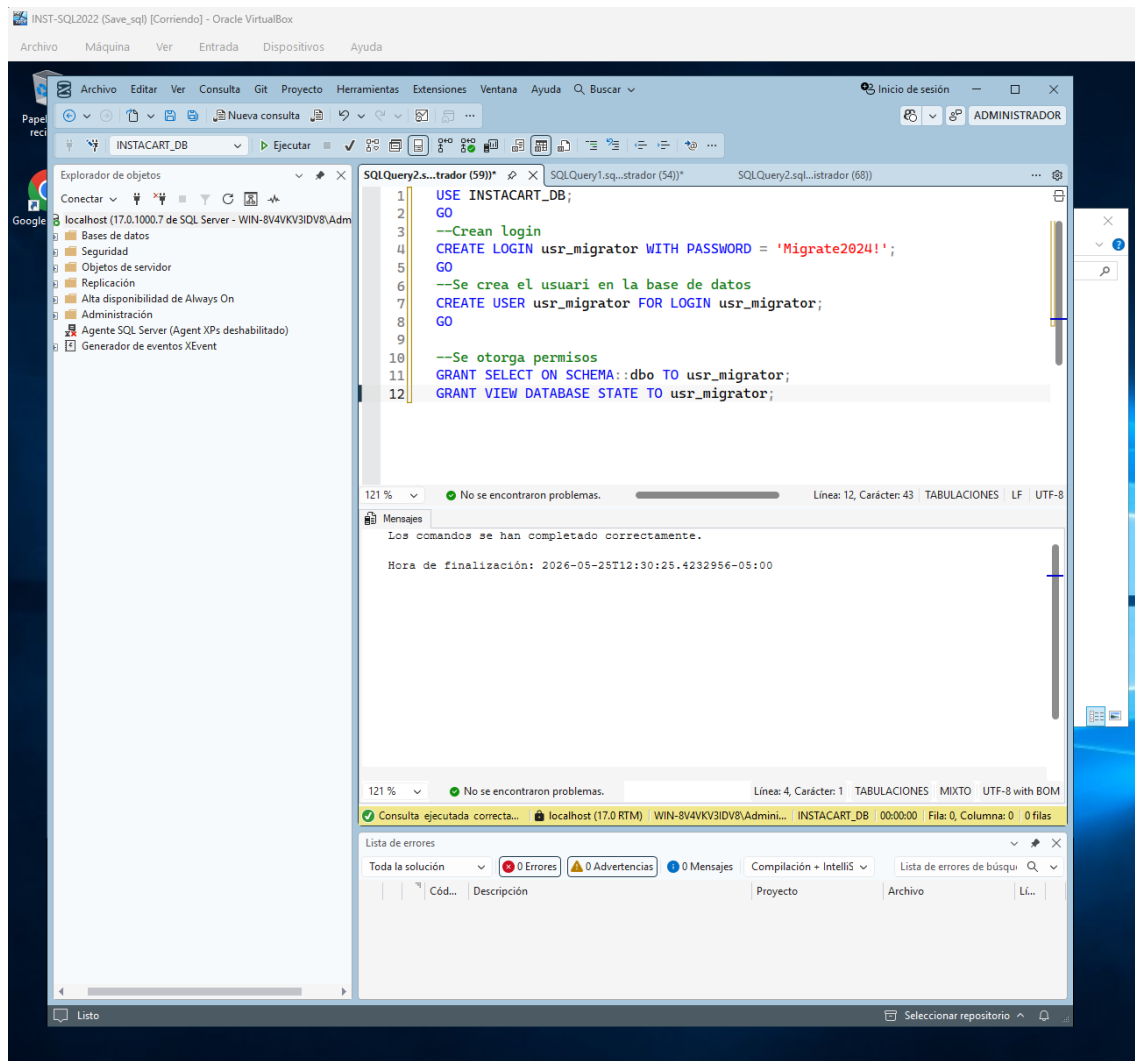
	products
1	49688



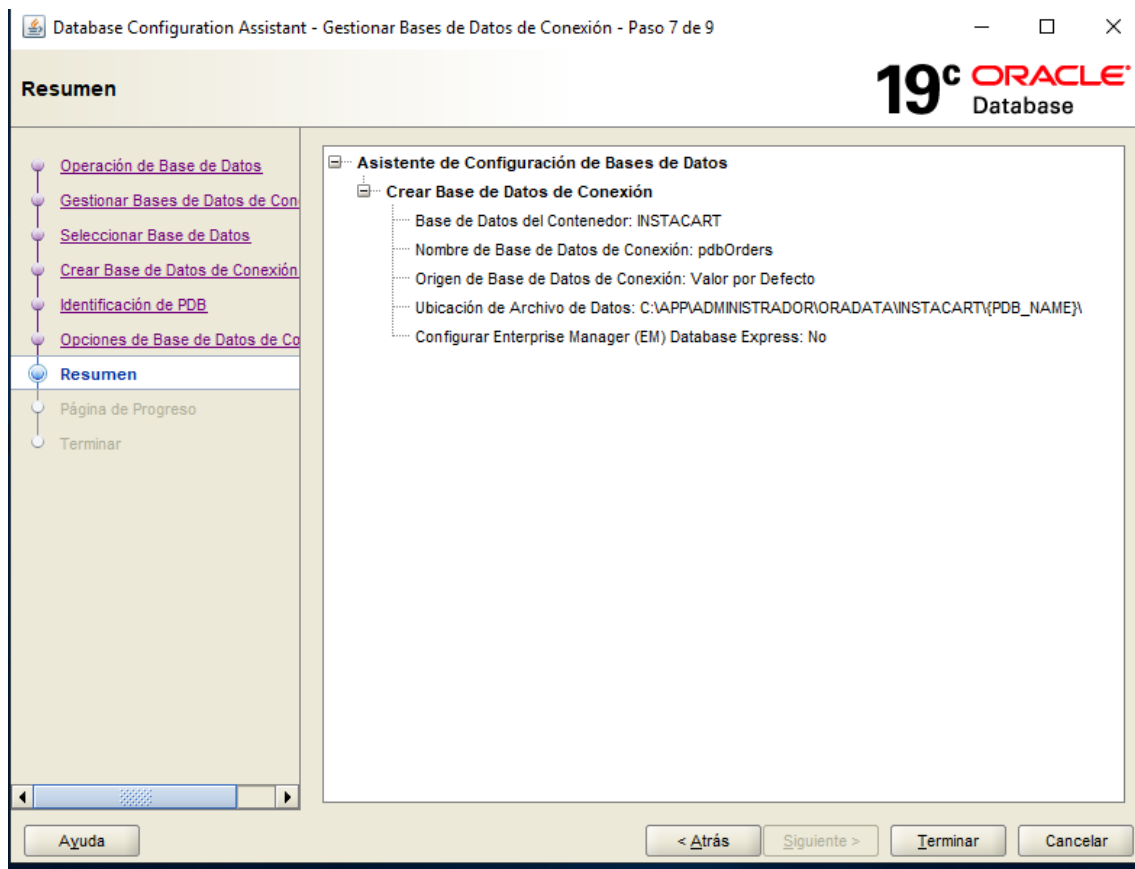
MOSTRANDO DATOS INGRESADOS POR EL BULK:



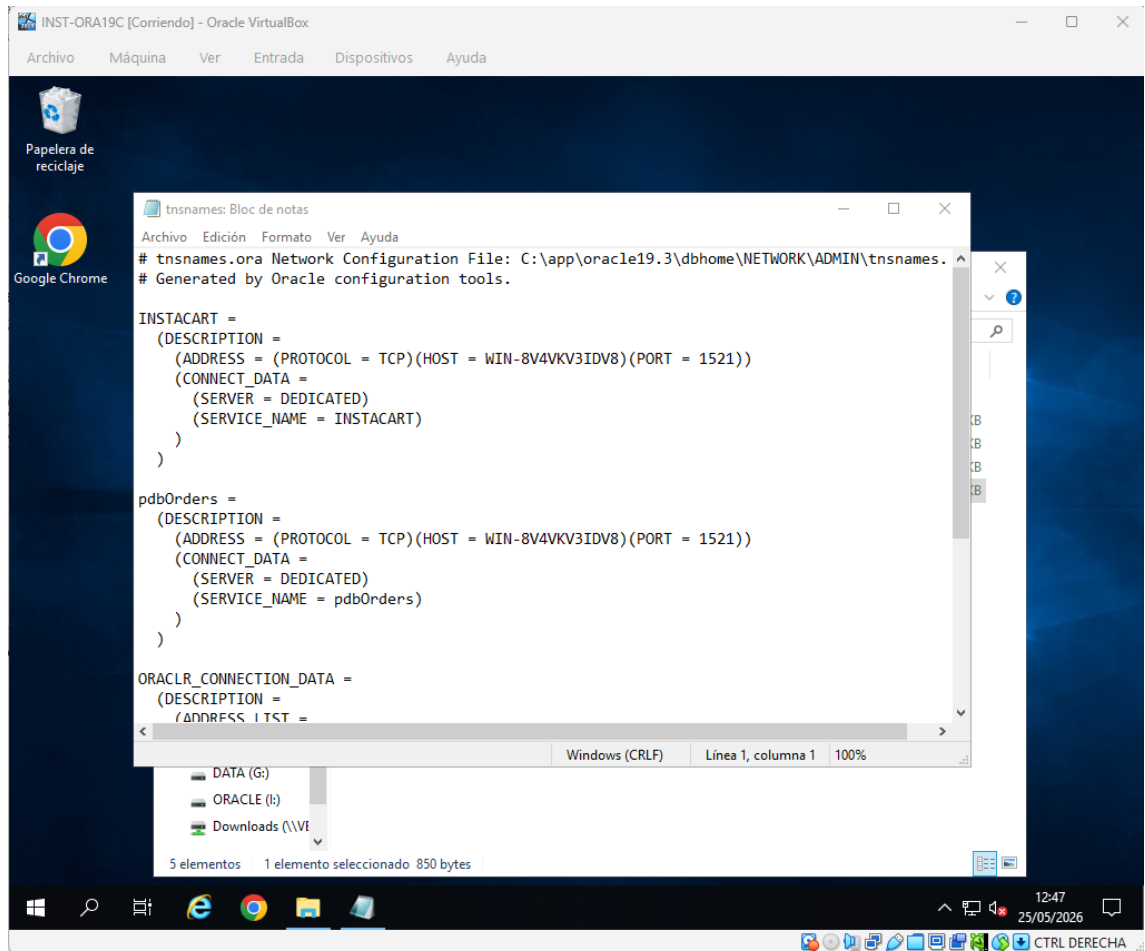
PERFIL DE USUARIO PARA LA MIGRACION DE DATOS EN SQL:



CREACION DE PDBORDERS EN CDB INSTACARD:



COFIGURACION DE TSNAME:



The screenshot shows a Windows VirtualBox environment with the title bar 'INST-ORA19C [Corriendo] - Oracle VirtualBox'. The desktop background is dark blue. On the left, there are icons for 'Papelera de reciclaje' and 'Google Chrome'. A 'tnsnames: Bloc de notas' window is open, displaying the following text:

```
# tnsnames.ora Network Configuration File: C:\app\oracle19.3\dbhome\NETWORK\ADMIN\tnsnames.
# Generated by Oracle configuration tools.

INSTACART =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-8V4VKV3IDV8)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = INSTACART)
    )
  )

pdbOrders =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-8V4VKV3IDV8)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = pdbOrders)
    )
  )

ORACLR_CONNECTION_DATA =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = WIN-8V4VKV3IDV8)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = ORACLR)
    )
  )
```

The status bar at the bottom of the window shows 'Linea 1, columna 1' and '100%'. The taskbar at the bottom of the screen shows the Windows logo, search icon, task view icon, and several application icons. The system clock in the bottom right corner shows '12:47' and '25/05/2026'.

MULTIPLEXANDO CONTROL FILES:

```
SQL> sho conn
SP2-0158: opci n SHOW desconocida "conn"
SQL> sho conn_name
SP2-0158: opci n SHOW desconocida "conn_name"
SQL> sho con_name

CON_NAME
-----
CDB$ROOT
SQL> sho pdba

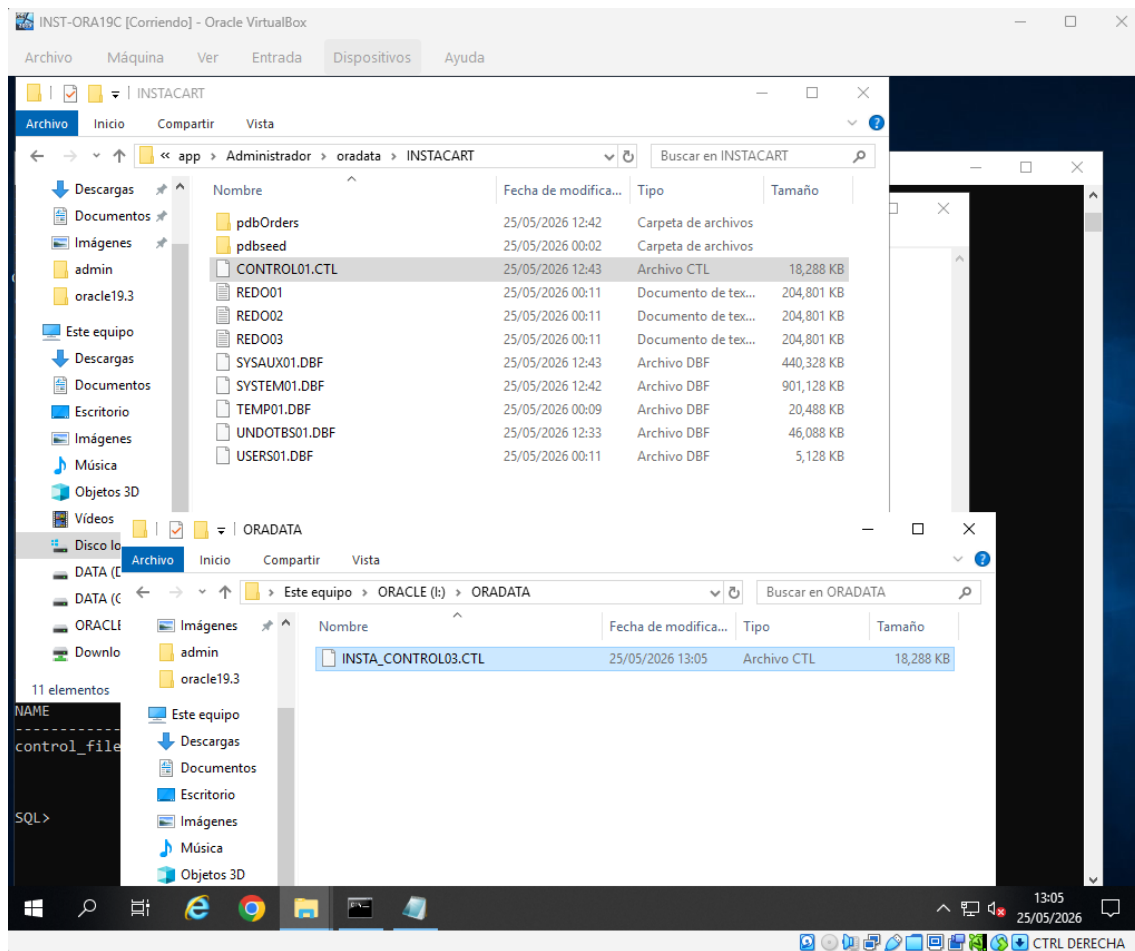
  CON_ID CON_NAME                                OPEN MODE RESTRICTED
-----
      2 PDB$SEED                                READ ONLY NO
      3 PDBORDERS                               READ WRITE NO
SQL> CONNECT sys/Admin123@PDBORDERS as sysdba
Conectado.
SQL> col name format a60
SQL> select status,name from v$controlfile;

STATUS NAME
-----
C:\APP\ADMINISTRADOR\ORADATA\INSTACART\CONTROL01.CTL
C:\APP\ADMINISTRADOR\FAST_RECOVERY_AREA\INSTACART\CONTROL02.
CTL

SQL> show parameter control_files

NAME                                TYPE      VALUE
-----
control_files                       string    C:\APP\ADMINISTRADOR\ORADATA\I
NSTACART\CONTROL01.CTL, C:\APP
\ADMINISTRADOR\FAST_RECOVERY_A
REA\INSTACART\CONTROL02.CTL

SQL>
```




```
INST-ORA19C [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

INSTACART
Archivo  Inicio  Compartir  Vista

Administrador: Símbolo del sistema - sqlplus / as sysdba

Instancia ORACLE iniciada.

Total System Global Area 1946155928 bytes
Fixed Size                 9268120 bytes
Variable Size             452984832 bytes
Database Buffers         1476395008 bytes
Redo Buffers              7507968 bytes
ORA-00214: el archivo de control
'C:\APP\ADMINISTRADOR\ORADATA\INSTACART\CONTROL01.CTL' version 1299 es
inconsistente con el archivo 'I:\ORADATA\INSTA_CONTROL03.CTL' version 1285

SQL> SHUTDOWN IMMEDIATE;
ORA-01507: base de datos sin montar

Instancia ORACLE cerrada.
SQL> STARTUP;
Instancia ORACLE iniciada.

Total System Global Area 1946155928 bytes
Fixed Size                 9268120 bytes
Variable Size             452984832 bytes
Database Buffers         1476395008 bytes
Redo Buffers              7507968 bytes
Base de datos montada.
Base de datos abierta.
SQL> col name format a60
SQL> select status,name from v$controlfile;

STATUS  NAME
-----
C:\APP\ADMINISTRADOR\ORADATA\INSTACART\CONTROL01.CTL
C:\APP\ADMINISTRADOR\FAST_RECOVERY_AREA\INSTACART\CONTROL02.
CTL

I:\ORADATA\INSTA_CONTROL03.CTL

SQL>
```

CREACION DE TABLESPACES EN ORACLE:

```
1
C:\APP\ADMINISTRADOR\ORADATA\INSTACART\REDO01.LOG

2
C:\APP\ADMINISTRADOR\ORADATA\INSTACART\REDO02.LOG

3
C:\APP\ADMINISTRADOR\ORADATA\INSTACART\REDO03.LOG

SQL> ALTER PLUGGABLE DATABASE PDBORDERS OPEN;
Base de datos de conexión modificada.

SQL> CONNECT sys/Admin2024!@PDBORDERS as sysdba
Conectado.
SQL> CREATE TABLESPACE tbsInstacart
2 DATAFILE 'G:\INSTACART\tbspaces\tbsInstacart01.dbf' SIZE 100M AUTOEXTEND ON NEXT 50M
3 EXTENT MANAGEMENT LOCAL AUTOALLOCATE
4 SEGMENT SPACE MANAGEMENT AUTO;

Tablespace creado.

SQL> CREATE TABLESPACE tbsIndex
2 DATAFILE 'I:\INSTACART\tbspaces\tbsIndex01.dbf' SIZE 50M AUTOEXTEND ON NEXT 10M
3 EXTENT MANAGEMENT LOCAL AUTOALLOCATE
4 SEGMENT SPACE MANAGEMENT AUTO;

Tablespace creado.

SQL>
```

CREACION DE USUARIO EN ORACLE PARA MIGRACION, DANDO PRIVILEGIOS:

```
Administrador: Símbolo del sistema - sqlplus / as sysdba
SQL> CREATE USER usr_migrator IDENTIFIED BY "Migrate2024!"
2  DEFAULT TABLESPACE tbsInstacart
3  TEMPORARY TABLESPACE TEMP
4  QUOTA UNLIMITED ON tbsInstacart
5  QUOTA UNLIMITED ON tbsIndex;

Usuario creado.

SQL> GRANT CREATE SESSION TO usr_migrator;

Concesión terminada correctamente.

SQL> GRANT CREATE TABLE TO usr_migrator;

Concesión terminada correctamente.

SQL> GRANT CREATE INDEX TO usr_migrator;
GRANT CREATE INDEX TO usr_migrator
*
ERROR en línea 1:
ORA-00990: falta el privilegio o no es válido

SQL> GRANT CREATE SEQUENCE TO usr_migrator;

Concesión terminada correctamente.

SQL> GRANT CREATE TRIGGER TO usr_migrator;

Concesión terminada correctamente.

SQL> GRANT CREATE DATABASE LINK TO usr_migrator;

Concesión terminada correctamente.

SQL> GRANT SELECT ANY TABLE TO usr_migrator;

Concesión terminada correctamente.
```

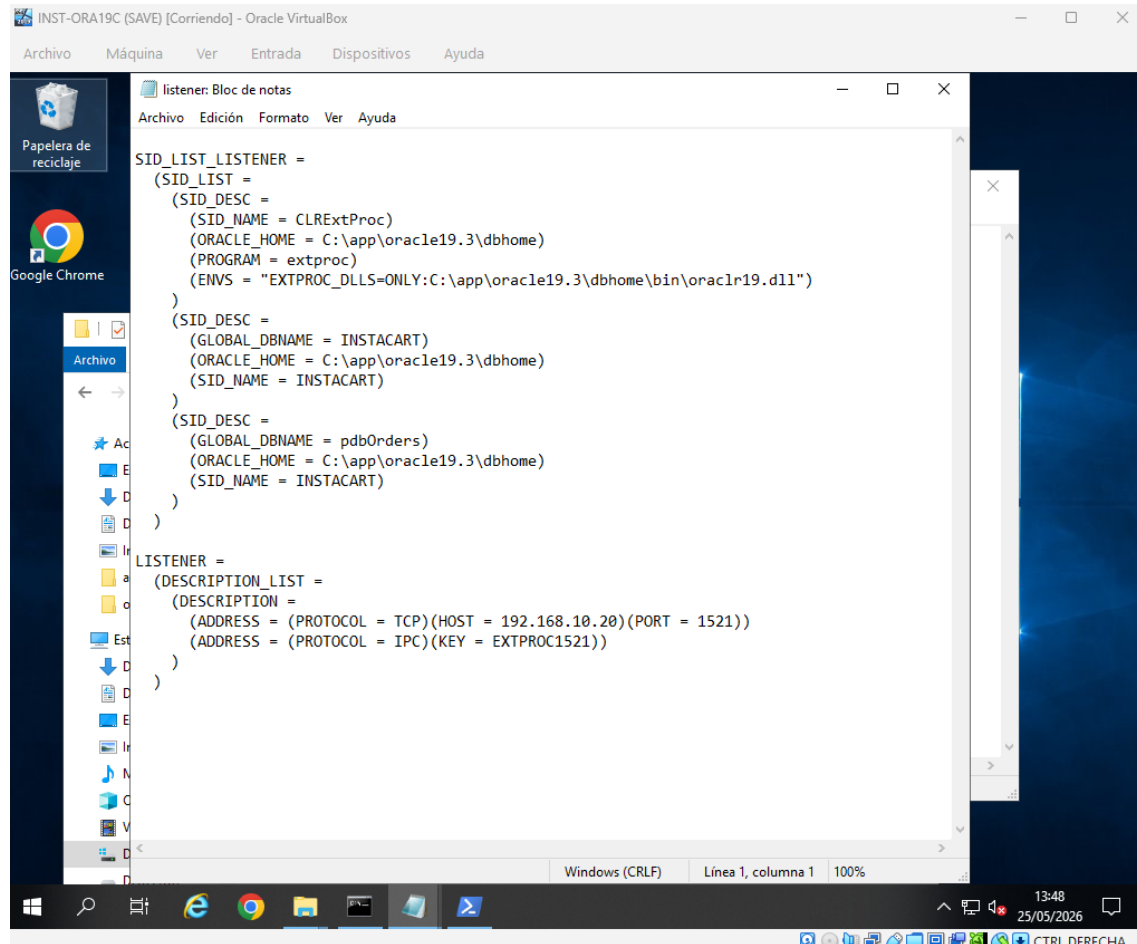
VERIFICAMOS:

```
SQL> SELECT username, default_tablespace, account_status FROM dba_users
2  WHERE username = 'USR_MIGRATOR';

USERNAME
-----
DEFAULT_TABLESPACE      ACCOUNT_STATUS
-----
USR_MIGRATOR
TBSINSTACART            OPEN

SQL>
```

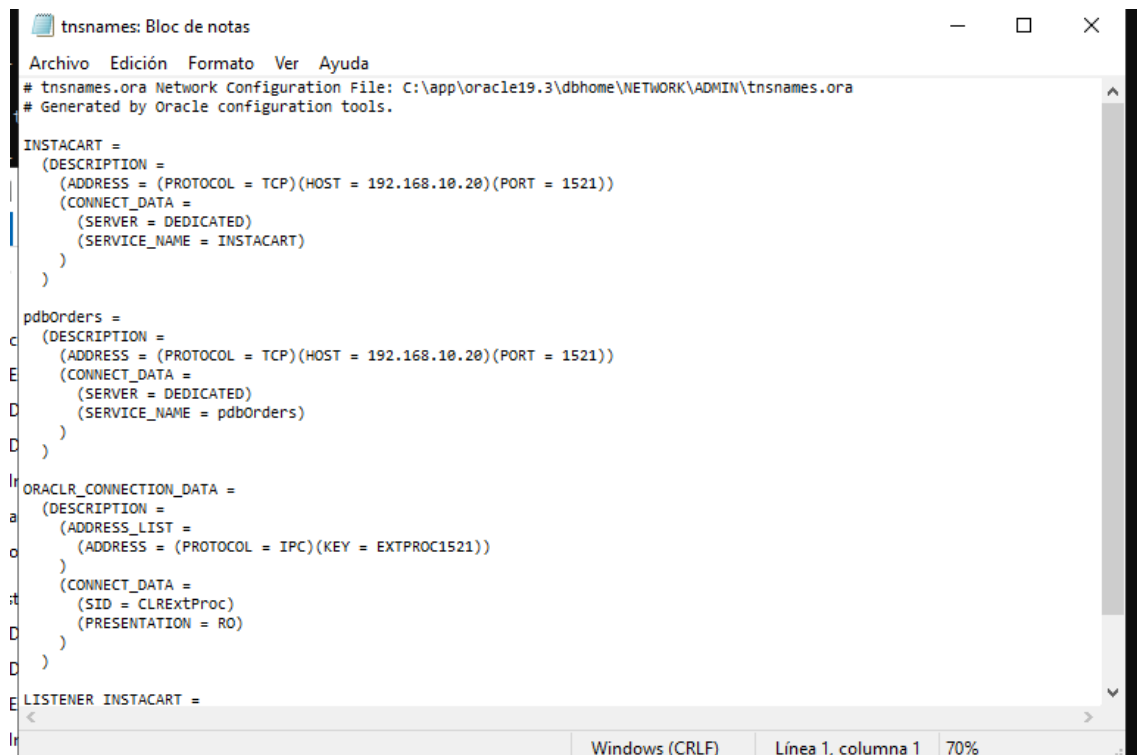
CONFIGURACION DEL LISTENER:



```
listener: Bloc de notas
Archivo Edición Formato Ver Ayuda

SID_LIST_LISTENER =
(SID_LIST =
(SID_DESC =
(SID_NAME = CLRExtProc)
(ORACLE_HOME = C:\app\oracle19.3\dbhome)
(PROGRAM = extproc)
(ENVS = "EXTPROC_DLLS=ONLY:C:\app\oracle19.3\dbhome\bin\oraclr19.dll")
)
(SID_DESC =
(GLOBAL_DBNAME = INSTACART)
(ORACLE_HOME = C:\app\oracle19.3\dbhome)
(SID_NAME = INSTACART)
)
(SID_DESC =
(GLOBAL_DBNAME = pdbOrders)
(ORACLE_HOME = C:\app\oracle19.3\dbhome)
(SID_NAME = INSTACART)
)
)
)
LISTENER =
(DESCRIPTION_LIST =
(DESCRIPTION =
(ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.10.20)(PORT = 1521))
(ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
)
)
)
```

CONFIGURACION DEL TSNAME:



```
tnsnames: Bloc de notas
Archivo Edición Formato Ver Ayuda

# tnsnames.ora Network Configuration File: C:\app\oracle19.3\dbhome\NETWORK\ADMIN\tnsnames.ora
# Generated by Oracle configuration tools.

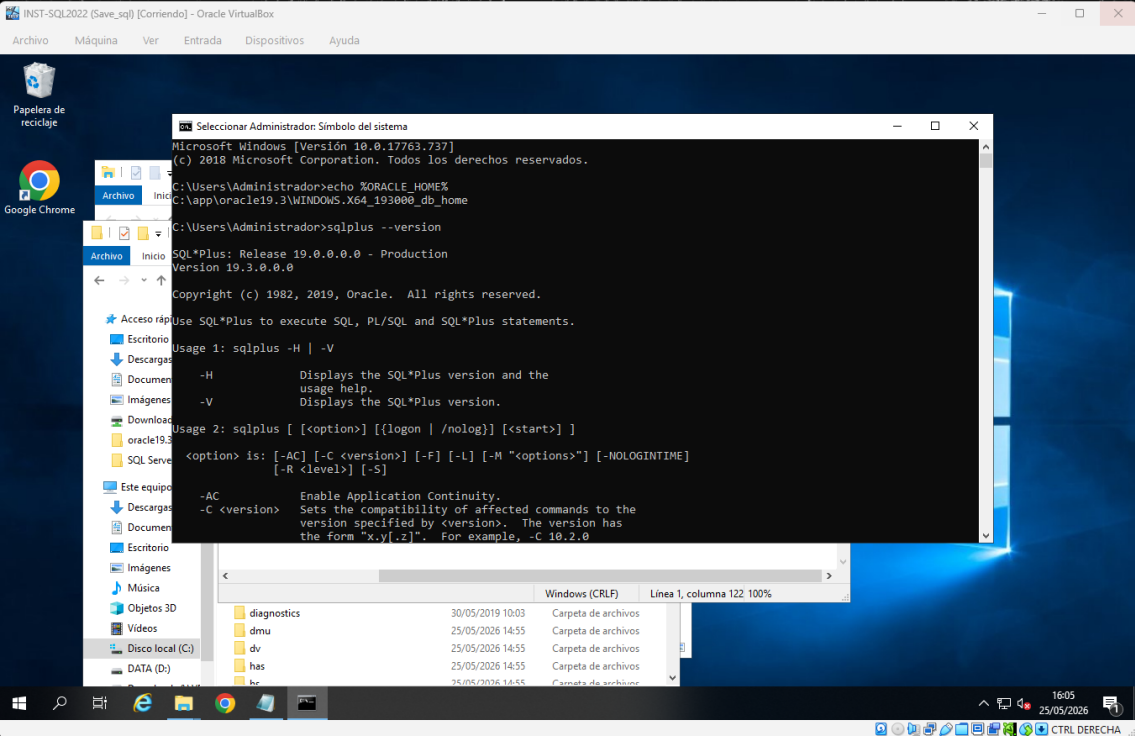
INSTACART =
(DESCRIPTION =
(ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.10.20)(PORT = 1521))
(CONNECT_DATA =
(SERVER = DEDICATED)
(SERVICE_NAME = INSTACART)
)
)
)

pdbOrders =
(DESCRIPTION =
(ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.10.20)(PORT = 1521))
(CONNECT_DATA =
(SERVER = DEDICATED)
(SERVICE_NAME = pdbOrders)
)
)
)

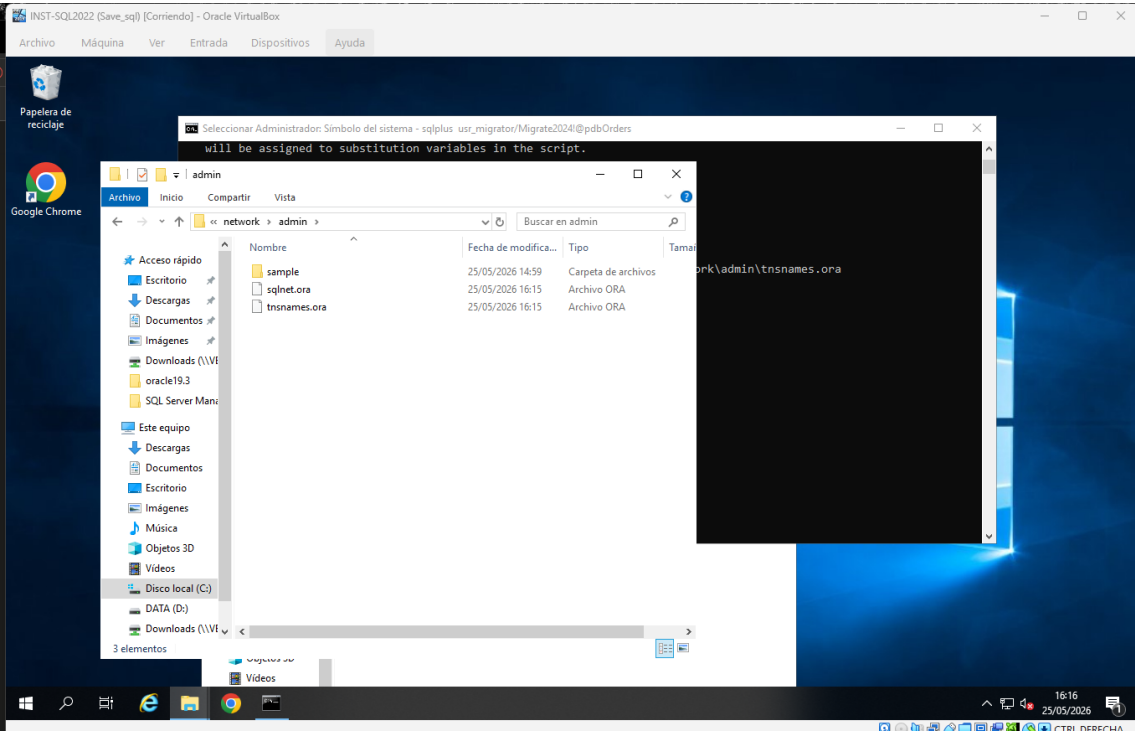
ORACLR_CONNECTION_DATA =
(DESCRIPTION =
(ADDRESS_LIST =
(ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
)
(CONNECT_DATA =
(SID = CLRExtProc)
(PRESENTATION = RO)
)
)
)

LISTENER INSTACART =
```

INSTALACION DE SQLPLUS EN INST SQL:



CREACIÓN DE TSNAME Y SQLNET:



CONEXIÓN DESDE SQL A ORACLE CON EL USR_MIGRATION Y EN LA PDBORDERS:

```
Administrador: Símbolo del sistema - sqlplus usr_migrator/Migrate2024!@pdbOrders
(e.g. $ORACLE_HOME/sqlplus/admin/glogin.sql) and the user profile
(e.g. login.sql in the working directory) are run. The files may
contain SQL*Plus commands.

Refer to the SQL*Plus User's Guide and Reference for more information.

C:\Users\Administrador>type C:\app\oracle19.3\WINDOWS.X64_193000_db_home\network\admin\tnsnames.ora
El sistema no puede encontrar el archivo especificado.

C:\Users\Administrador>sqlplus usr_migrator/Migrate2024!@pdbOrders

SQL*Plus: Release 19.0.0.0.0 - Production on Mon May 25 16:16:07 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

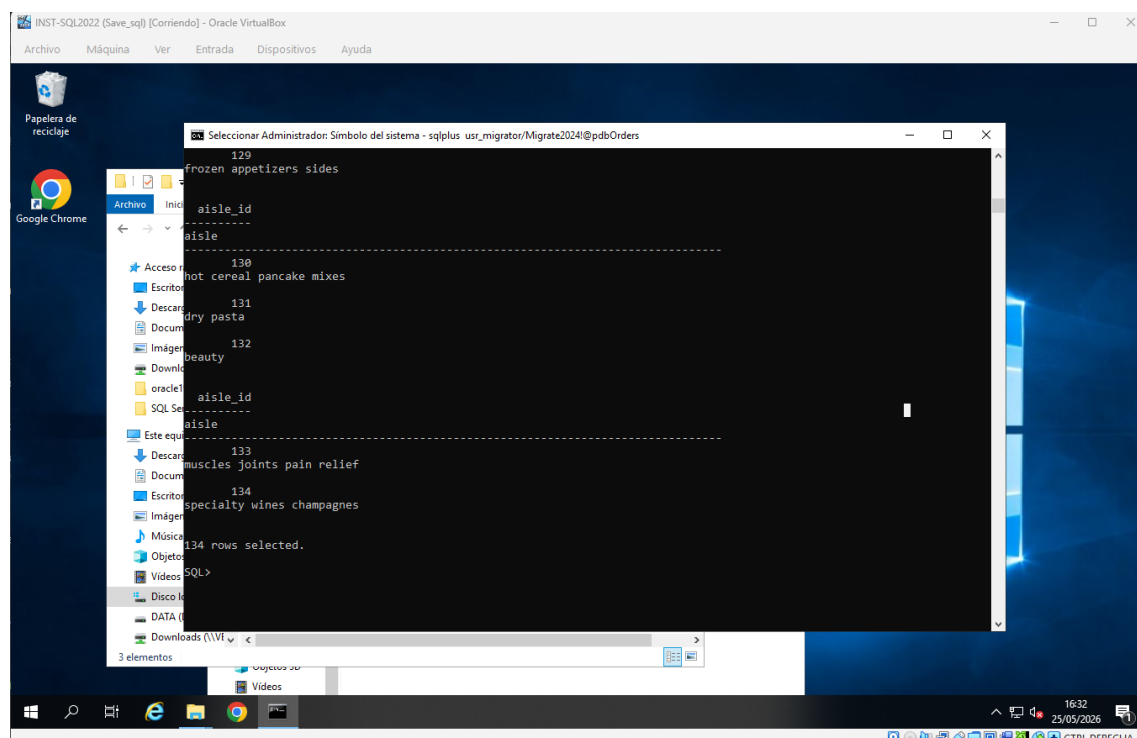
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> SHOW CON_NAME;

CON_NAME
-----
PDBORDERS
SQL> SELECT username FROM user_users;

USERNAME
-----
USR_MIGRATOR

SQL>
```



CREANDO TABLAS EN ORACLE Y TRANSFIRIENDO DATOS:

1.- AISLES

```
Seleccionar Administrador: Símbolo del sistema - sqlplus usr_migrator/Migrate2024!@pdbOrders
beauty

  aisle_id
-----
aisle
-----
      133
muscles joints pain relief
      134
specialty wines champagnes

134 rows selected.

SQL> CREATE TABLE aisles (
  2   aisle_id INT NOT NULL PRIMARY KEY,
  3   aisle VARCHAR2(100) NOT NULL
  4 ) TABLESPACE tbsInstacart;

Table created.

SQL>
SQL> INSERT INTO aisles SELECT * FROM "dbo"."aisles"@sqlserver_link;

134 rows created.

SQL> COMMIT;

Commit complete.

SQL> _
```

2.- DEPARTMENTS:

```
SQL>
SQL> INSERT INTO aisles SELECT * FROM "dbo"."aisles"@sqlserver_link;

134 rows created.

SQL> COMMIT;

Commit complete.

SQL>
SQL> CREATE TABLE departments (
  2   department_id INT NOT NULL PRIMARY KEY,
  3   department VARCHAR2(100) NOT NULL
  4 ) TABLESPACE tbsInstacart;

Table created.

SQL> INSERT INTO departments SELECT * FROM "dbo"."departments"@sqlserver_link;

21 rows created.

SQL> COMMIT;

Commit complete.

SQL>
```

3.- PRODUCTOS

```
SQL> CREATE TABLE products (  
  2   product_id INT NOT NULL PRIMARY KEY,  
  3   product_name VARCHAR2(500) NOT NULL,  
  4   aisle_id INT,  
  5   department_id INT  
  6 ) TABLESPACE tbsInstacart;  
  
Table created.  
  
SQL>  
SQL> INSERT INTO products SELECT * FROM "dbo"."products"@sqlserver_link;  
  
49688 rows created.  
  
SQL> COMMIT;
```

4.- ORDERS

```
Table created.  
  
SQL>  
SQL> INSERT INTO orders SELECT * FROM "dbo"."orders"@sqlserver_link;  
  
3421083 rows created.  
  
SQL> COMMIT;  
  
Commit complete.  
  
SQL>
```

5.- ORDER PRODUCTS

```
INST-SQL2022 (Save_sql) [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

admin
Administrador: Símbolo del sistema - sqlplus usr_migrator/Migrate2024/@pdbOrders

order_id hasta 600000 | total: 5042083
order_id hasta 700000 | total: 6032457
order_id hasta 800000 | total: 7019827
order_id hasta 900000 | total: 8007061
order_id hasta 1000000 | total: 8994508
order_id hasta 1100000 | total: 9980874
order_id hasta 1200000 | total: 10967862
order_id hasta 1300000 | total: 11958270
order_id hasta 1400000 | total: 12948271
order_id hasta 1500000 | total: 13936467
order_id hasta 1600000 | total: 14926596
order_id hasta 1700000 | total: 15917084
order_id hasta 1800000 | total: 16908030
order_id hasta 1900000 | total: 17897483
order_id hasta 2000000 | total: 18887096
order_id hasta 2100000 | total: 19873674
order_id hasta 2200000 | total: 20859959
order_id hasta 2300000 | total: 21850089
order_id hasta 2400000 | total: 22839790
order_id hasta 2500000 | total: 23828206
order_id hasta 2600000 | total: 24816916
order_id hasta 2700000 | total: 25807042
order_id hasta 2800000 | total: 26799275
order_id hasta 2900000 | total: 27787786
order_id hasta 3000000 | total: 28773910
order_id hasta 3100000 | total: 29760153
order_id hasta 3200000 | total: 30744151
order_id hasta 3300000 | total: 31735582
order_id hasta 3400000 | total: 32722250
order_id hasta 3500000 | total: 32931847
COMPLETADO. Total: 32931847

PL/SQL procedure successfully completed.

SQL>
```

VERIFICAMOS:

```
order_id hasta 3500000 | total: 32931847
COMPLETADO. Total: 32931847

PL/SQL procedure successfully completed.

SQL> SELECT COUNT(*) FROM order_products;

COUNT(*)
-----
32931847

SQL>
```


Y VEMOS LAS TABALAS JUNTAS PARA VERIFICAR LA MIGRACION:

```
-----  
32931847  
  
SQL> SELECT 'aisles' AS tabla, COUNT(*) AS registros FROM aisles UNION ALL  
2  SELECT 'departments', COUNT(*) FROM departments UNION ALL  
3  SELECT 'products', COUNT(*) FROM products UNION ALL  
4  SELECT 'orders', COUNT(*) FROM orders UNION ALL  
5  SELECT 'order_products', COUNT(*) FROM order_products;
```

TABLA	REGISTROS
aisles	134
departments	21
products	49688
orders	3421083
order_products	32931847

```
SQL> █
```